

③ Manipulator Kinematics

Recall

rigid body motion form a 16 group of $SE(3)$

$$g = (R, p) \in SE(3)$$

$\swarrow$
R rotational matrix
p translation vector

§ 1. Forward Kinematics

Note

6 lower pairs of joint mechanics

revolute, prismatic, helical, cylindrical, spherical, planar

Set up

• two joints revolute $\theta \in S^1 = [0, 2\pi)$ (rotation)

prismatic $q \in \mathbb{R}$ (translation)

• joint space $Q = \mathbb{T}^p \times \mathbb{R}^r$ where $\mathbb{T}^p = S^1 \times S^1 \times \dots \times S^1$ (p-torus)

$\swarrow$
p revolute r prismatic joints

ex. JCARA $Q = \mathbb{T}^3 \times \mathbb{R}$

• coordinate frames S = base frame (fixed)

T = tool frame (end effector)

Problem

Given joint angles $q \in Q$, find configuration of T relative to S

$$g_{st} : Q \rightarrow SE(3)$$

Recall

Build g_{st} by composing transformations between adjacent links

(Denavit

-Hartenberg)

$$S \xrightarrow{\theta_1} L_1 \xrightarrow{\theta_2} L_2 \xrightarrow{\theta_3} \dots \xrightarrow{\theta_n} L_n \rightarrow T$$

$$\rightarrow g_{st}(\theta) = g_{sL_1}(\theta_1) g_{L_1 L_2}(\theta_2) \dots g_{L_{n-1} L_n}(\theta_n) g_{L_n T}$$

where $g_{i-1,i}(\theta_i)$ is $SE(3)$ transformation

$\Downarrow$

requires n different coordinate systems to set up & track

(fiddly, error-prone)

§ 1.1 Product of Exponentials formula

Recall each joint generates a screw motion $g_{ab}(\theta) = e^{\hat{E}_a \theta} g_{ab}(0)$

Idea If E_i is twist associated with joint axis, moving through angle θ simply left multiplies the reference configuration by $e^{\hat{E}_i \theta}$

Derivation (2 DOF)

- fix $\theta_1 = 0$ move only θ_2 $g_{12}(\theta_2) = e^{\hat{E}_2 \theta_2} g_{12}(0)$
- fix θ_2 and move θ_1 $g_{12}(\theta_1, \theta_2) = e^{\hat{E}_1 \theta_1} g_{12}(\theta_2)$
- $\Rightarrow g_{12}(\theta_1, \theta_2) = e^{\hat{E}_1 \theta_1} e^{\hat{E}_2 \theta_2} g_{12}(0)$

Note E_i is a constant twist evaluated at $\theta = 0$ so no need to track moving frame.

Claim PoE does not depend on the order

(proof) construct via moving θ_1 first

- fix $\theta_2 = 0$ move θ_1 $g_{12}(\theta_1) = e^{\hat{E}_1 \theta_1} g_{12}(0)$
- move θ_2 (since joint 1 moved the entire arm, joint 2 no longer E_2)
let new axis be E'_2

$E'_2 = \text{Ad}_{e^{\hat{E}_1 \theta_1}} E_2$ so motion due to θ_2 about this axis is
 $e^{\hat{E}'_2 \theta_2} = e^{(\text{Ad}_{e^{\hat{E}_1 \theta_1}} E_2) \theta_2}$

recall from chapter 2, $g e^{\hat{X} \theta} g^{-1} = e^{(\text{Ad}_g X) \theta}$

Hence, $e^{\hat{E}'_2 \theta_2} = e^{\hat{E}_1 \theta_1} e^{\hat{E}_2 \theta_2} e^{-\hat{E}_1 \theta_1}$

composing both motions,

$$\begin{aligned} g_{12}(\theta_1, \theta_2) &= e^{\hat{E}'_2 \theta_2} e^{\hat{E}_1 \theta_1} g_{12}(0) \\ &= e^{\hat{E}_1 \theta_1} e^{\hat{E}_2 \theta_2} \underbrace{e^{-\hat{E}_1 \theta_1} e^{\hat{E}_1 \theta_1}}_I g_{12}(0) = e^{\hat{E}_1 \theta_1} e^{\hat{E}_2 \theta_2} g_{12}(0) \end{aligned}$$

PoE

Extending to n joints by induction,

$$g_{1n}(\theta) = e^{\hat{E}_1 \theta_1} e^{\hat{E}_2 \theta_2} \dots e^{\hat{E}_n \theta_n} g_{1n}(0)$$

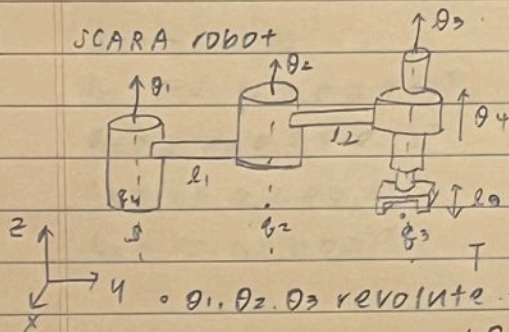
RemarkHow to build the twist ξ_i

• revolute joint τ $\xi_i = \begin{pmatrix} -w_i \times q_i \\ w_i \end{pmatrix}$ $w_i \in \mathbb{R}^3$ unit axis direction
 $q_i \in \mathbb{R}^3$ any point of axis

• prismatic joint τ $\xi_i = \begin{pmatrix} v_i \\ 0 \end{pmatrix}$ $v_i \in \mathbb{R}^3$ unit translation direction
 $\hookrightarrow w_i, q_i, v_i$ measured in base frame, at reference config $\theta = 0$

Example

SCARA robot



reference frame

$$g_{1,2}(0) = \begin{pmatrix} I & p_0 \\ 0 & 1 \end{pmatrix}, \quad p_0 = \begin{pmatrix} 0 \\ l_1 + l_2 \\ l_0 \end{pmatrix}$$

• $\theta_1, \theta_2, \theta_3$ revolute - pointing in z direction

$$w_1 = w_2 = w_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

points on each axis at $\theta = 0$ $q_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ $q_2 = \begin{pmatrix} 0 \\ l_1 \\ 0 \end{pmatrix}$ $q_3 = \begin{pmatrix} l_1 + l_2 \\ 0 \\ 0 \end{pmatrix}$ now build twist using $\begin{pmatrix} -w_i \times q_i \\ w_i \end{pmatrix}$

$$\xi_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad \xi_2 = \begin{pmatrix} l_1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \xi_3 = \begin{pmatrix} l_1 + l_2 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

• θ_4 prismatic sliding in z direction

$$\xi_4 = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad v_4 = (0, 0, -1)^T$$

 $\Rightarrow$ forward kinematic map T $g_{1,2}(\theta) = e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} e^{\hat{\xi}_3 \theta_3} e^{\hat{\xi}_4 \theta_4} g_{1,2}(0)$

$$= \begin{bmatrix} R(\theta) & p(\theta) \\ 0 & 1 \end{bmatrix}$$

$$\text{where } R(\theta) = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$p(\theta) = \begin{bmatrix} -l_1 \sin \theta_1 - l_2 \sin(\theta_1 + \theta_2) \\ l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2) \\ l_0 + \theta_4 \end{bmatrix}$$

* can read geometry directly from formula

§1.2 parameterization of manipulators via twists

Alternative J coincident with T at $\theta=0$: $g_{J,0}(0)=I$
 $\therefore g_{J,0}(\theta) = e^{\xi_1 \theta_1} e^{\xi_2 \theta_2} \dots e^{\xi_n \theta_n}$

Example SCARA

$$\xi_1 = (-l_1 - l_2, 0, 0)^T$$

$$\xi_2 = (-l_2, 0, 0)^T$$

$$\xi_3 = (-l_1 - l_2, 0, 0, 0, 0, 1)^T$$

$$\xi_4 = (-l_2, 0, 0, 0, 0, 1)^T$$

$$\xi_5 = (0, 0, 0, 0, 0, 1)^T$$

$$\xi_6 = (0, 1, 0, 0, 0, 0)^T$$

same robot, different frame,
different twist

Remark

DH vs POE

DH $\rightarrow$ assigns coordinate frame to every single link ($n+2$ frames)
 with local frame-dependent representation.

POE $\rightarrow$ 2 frames (S.T) with global geometric representation.

conversion formula DH $\rightarrow$ POE

$\xi_i = \text{Ad}_{g_{e, i-1}(0)} \xi_{i+1, T}$ take local DH twist and push to base frame S
 using adjoint of reference config. of frame e_{i-1}

§1.3. Manipulator workspace

Question what is set of all configurations the end effector can actually reach?

Def

complete workspace $W = \{g_{J,0}(\theta) : \theta \in Q\} \subset SE(3)$

Reachable workspace $W_R = \{p(\theta) : \theta \in Q\} \subset \mathbb{R}^3$

Dextrous workspace $W_D = \{p \in \mathbb{R}^3 : \forall R \in SO(3), \exists \theta \text{ with } g_{J,0}(\theta) = (p, R) \in \mathbb{R}^3 \times \mathbb{R}^3\}$

$\rightarrow W_D \subseteq W_R \subseteq W$

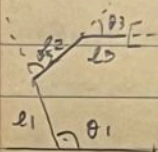
DefSpherical wrist

Three revolute joints whose axes all intersect at a single point
 spherical wrist + end-effector frame at wrist centre:

$$W_D = W_R \text{ and } W = W_R \text{JD}(\theta)$$

Remark

spherical wrist can achieve any orientation $\rightarrow$ most industrial robot's design

Exampleplanar arm

forward kinematics give

$$x = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2) + l_3 \cos(\theta_1 + \theta_2 + \theta_3)$$

$$y = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2) + l_3 \sin(\theta_1 + \theta_2 + \theta_3)$$

to find W_R , ignore orientation & build reachable set

- 1) fix θ_1, θ_2 sweep $\theta_3 \in [0, 2\pi] \rightarrow$ circle of radius l_3
- 2) fix θ_1 sweep $\theta_2 \in [0, 2\pi] \rightarrow$ inner $l_2 - l_3$ outer $l_2 + l_3$
- 3) sweep $\theta_1 \in [0, 2\pi] \rightarrow$ inner $l_1 - l_2 - l_3$ outer $l_1 + l_2 + l_3$ (W_R)

$W_D =$ annulus W_R shrunken by $\pm l_3$ on each edge

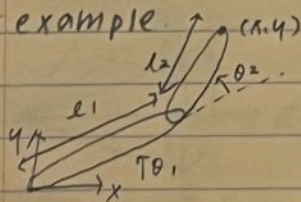
$\hookrightarrow$ inner $l_1 - l_2 + l_3$ outer $l_1 + l_2 - l_3$

so $W_D \subseteq W_R$ (position reachable at some angle but not every)

§ 2. Inverse kinematicsproblem

given $g_d \in SE(3)$, find $\theta \in \mathbb{R}$ s.t. $g_{\text{ic}}(\theta) = g_d$

(may have 0, 1 or multiple solutions)

§ 2.1 planar example

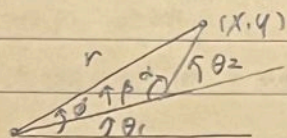
Forward kinematics

$$x = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2)$$

$$y = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2)$$

solution

use polar coordinates

by LAW OF COSINES can find θ_2

$$\theta_2 = \pi \pm \alpha, \quad \alpha = \cos^{-1} \left(\frac{l_1^2 + l_2^2 - r^2}{2l_1 l_2} \right)$$

↳ gives two solutions.

$$\text{then } \theta_1 = \text{atan2}(y, x) \pm \beta, \quad \beta = \cos^{-1} \left(\frac{r^2 + l_1^2 - l_2^2}{2l_1 r} \right)$$

0 solutions $r > l_1 + l_2$ (outside workspace)

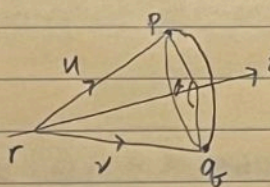
2 solutions generic (elbow up/down)

1 solution $d=0$ (fully extended/folded)§ 2.2 Paden Kahan subproblemsIdea

decompose into small geometric subproblems - each directly tied to PoE structure

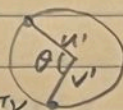
Type 1

Rotation about a single axis

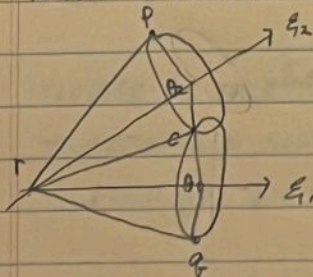
Given zero pitch unit twist ξ_1 , points $p, q \in \mathbb{R}^3$ find θ s.t. $e^{\xi_1 \theta} p = q$ 

→ project onto plane

$$u' = u - ww^T u, \quad v' = v - ww^T v$$

→ has a solution only if $w^T u = w^T v$ and $\|u\| = \|v\|$ → solution $\theta = \text{atan2}(w^T(u' \times v'), u'^T v')$ → 0 or 1 sol.Type 2

Rotation about two subsequent axes



Given intersecting zero pitch unit twists

 ξ_1, ξ_2 , points $p, q \in \mathbb{R}^3$, find θ_1, θ_2 s.t.

$$e^{\xi_1 \theta_1} e^{\xi_2 \theta_2} p = q$$

$$\rightarrow e^{\xi_2 \theta_2} p = c = e^{-\xi_1 \theta_1} q$$

Find c (circle-circle intersection)

$$\text{let } u = p - r, \quad v = q - r, \quad z = c - r$$

$$z = \alpha w_1 + \beta w_2 + \gamma (w_1 \times w_2)$$

$$\text{solve for } \alpha, \beta \text{ from projections then } \gamma^2 = \frac{\|u\|^2 - \alpha^2 - \beta^2 - 2\alpha\beta w_1^T w_2}{\|w_1 \times w_2\|^2}$$

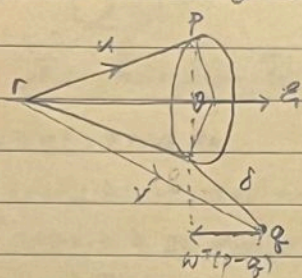
Find c .

Then $e^{\hat{E}_2 \theta_2} p = c$, $e^{\hat{E}_1 \theta_1} c = c$ is Type 1.

of sol: 0, 1 or 2 (depending on δ^2)

Type 3

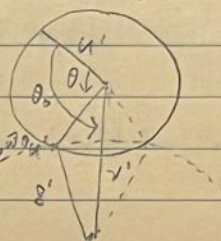
Rotation to a given distance.



Given zero pitch unit twist E_i , $p, q \in \mathbb{R}^3$

scalar $\delta > 0$, find θ s.t. $\|q - e^{\hat{E}_2 \theta} p\| = \delta$.

$\Rightarrow$ projection on plane



$$\delta^2 = \delta'^2 - \|w^T(p - q)\|^2$$

solution is

$$\theta = \theta_0 \pm \cos^{-1} \left(\frac{\|u'\|^2 + \|v'\|^2 - \delta'^2}{2\|u'\|\|v'\|} \right)$$

of sol: 0, 1 or 2 (depending on # points in which circle of radius

$\|u'\|$ intersects the circle of radius d')

§ 2.3 solving inverse kinematics using subproblems

TRICK 1

Apply both sides to a point p on the axis of joint 1

$$e^{\hat{E}_1 \theta_1} p = p \Rightarrow \theta_1 \text{ drops out}$$

TRICK 2

subtract a fixed point q from both sides and take norms

$$\|gp - q\| = \|e^{\hat{E}_1 \theta_1} e^{\hat{E}_2 \theta_2} e^{\hat{E}_3 \theta_3} p - q\| \Rightarrow \text{can eliminate unknowns}$$

EXAMPLE

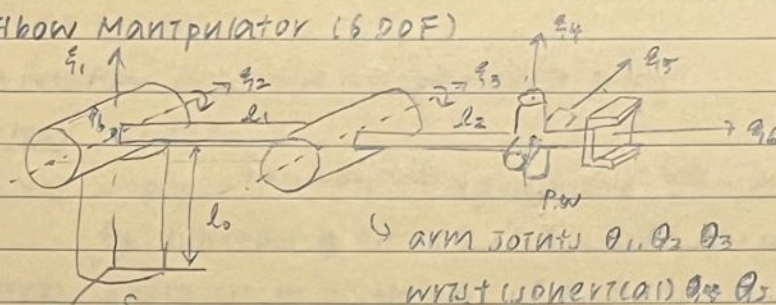
Isolate θ_3 in $e^{\hat{E}_1 \theta_1} e^{\hat{E}_2 \theta_2} e^{\hat{E}_3 \theta_3} = g$ where E_1, E_2 intersect at q

$\Rightarrow$ apply to p , subtract q , take norm

$$\begin{aligned} \delta &= \|gp - q\| = \|e^{\hat{E}_1 \theta_1} e^{\hat{E}_2 \theta_2} (e^{\hat{E}_3 \theta_3} p - q)\| \\ &= \|e^{\hat{E}_3 \theta_3} p - q\| \end{aligned}$$

Example

Elbow Manipulator (6 DOF)

ARM JOINTS $\theta_1, \theta_2, \theta_3$ WRIST (SPHERICAL) $\theta_4, \theta_5, \theta_6$

$$e^{\hat{z}_1 \theta_1} e^{\hat{z}_2 \theta_2} e^{\hat{z}_3 \theta_3} e^{\hat{z}_4 \theta_4} e^{\hat{z}_5 \theta_5} e^{\hat{z}_6 \theta_6} = g_d g_{it}(0) = g_1$$

step 1: Find θ_3 .apply both sides to p_w . (TRICK 1) $\rightarrow e^{\hat{z}_1 \theta_1} e^{\hat{z}_2 \theta_2} e^{\hat{z}_3 \theta_3} p_w = g_1 p_w$ (*)subtract p_w from both sides & take norm (TRICK 2) $\rightarrow$

$$\|e^{\hat{z}_3 \theta_3} p_w - p_w\| = \|g_1 p_w - p_w\| = d$$

 $\Rightarrow$ subproblem 3 with $p = p_w$, $q = p_w$, $d = \|g_1 p_w - p_w\|$ gives 0, 1 or 2 values of θ_3 .step 2 Find θ_1, θ_2

$$p' = e^{\hat{z}_3 \theta_3} p_w \text{ is known}$$

so (*) becomes $e^{\hat{z}_1 \theta_1} e^{\hat{z}_2 \theta_2} p' = g_1 p_w$ $\Rightarrow$ subproblem 2 with axes z_1, z_2 (intersect at p_w)gives 0, 1, or 2 pairs (θ_1, θ_2) step 3 Find θ_4, θ_5

move known joints to rhs to isolate wrist

$$e^{\hat{z}_4 \theta_4} e^{\hat{z}_5 \theta_5} e^{\hat{z}_6 \theta_6} = e^{-\hat{z}_1 \theta_1} e^{-\hat{z}_2 \theta_2} e^{-\hat{z}_3 \theta_3} g_d g_{it}(0) = g_2$$

Apply to point p on axis of z_6 but not z_4, z_5 (TRICK 1)

$$\text{Then } e^{\hat{z}_4 \theta_4} e^{\hat{z}_5 \theta_5} p = g_2 p$$

 $\Rightarrow$ subproblem 2 with axes z_4, z_5 gives 0, 1 or 2 pairs of (θ_4, θ_5) step 4 Find θ_6

Rearrange with all other angles known

 $\Rightarrow$ subproblem 1

unique solution if exists

Total up to 8 solutions

Example

SCARA (4 DOF)

→ revolute axes and one prismatic joint

step 1 $\theta_4 = z - z_0$ step 2 apply $e^{\hat{z}_1 \theta_1} e^{\hat{z}_2 \theta_2} = e^{\hat{z}_3 \theta_3} = g_3 g_4^{-1}(0) e^{-\hat{z}_4 \theta_4} = g_1$ to p on axis $\hat{z}_3$, subtract g_3 on axis of $\hat{z}_1$, take norm $\Rightarrow J P_3 \Rightarrow \theta_2$ step 3 apply (0) to p' on axis of $\hat{z}_3 \Rightarrow J P_1 \Rightarrow \theta_1$ step 4 rearrange for θ_3 alone $\Rightarrow J P_1 \Rightarrow \theta_3$ Total solutions: $J P_3 \times J P_1 \times J P_1 = 2 \times 1 \times 1 = 2$ § 2.4 General solutions to inverse kinematics problemif no interesting axes $\Rightarrow$ Paden Kahan not applicable. $\Rightarrow$ use analytical eliminationIdeasystematic elimination of $n-1$ variables from n polynomial equations $\Rightarrow$ single polynomial in one variablestep 1 convert to polynomial system.set $x_1 = \tan(\theta_1/2) \Rightarrow J_1 = J \tan \theta_1 = \frac{2x_1}{1+x_1^2}$ $c_1 = \cos \theta_1 = \frac{1-x_1^2}{1+x_1^2}$ $\Rightarrow g_1(\theta) = g_d$ becomes system of polynomial equations in $x_1, \dots, x_n$ step 2 analytical eliminationGiven n polynomial eq. $f_1 = \dots = f_n = 0$ in n unknowns $x_1, \dots, x_n$

- rewrite as linear system in monomials of $x_2, \dots, x_n$ with coefficients that are polynomials in x_1

$$A(x_1) m(x_2, \dots, x_n) = 0$$

- multiply existing eq. by monomials to generate new independent eq. until the system is square
- For non trivial sol to exist $\det A(x_1) = 0$

 $\Rightarrow$ yields $x_3 = \tan(\theta_3/2)$ (degree 16 polynomial)